

Basile Gros

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PhD Student @ VERIMAG

Academic Positions

2023 - Now **PhD in Computer Science**
supervised by **Pierre Corbineau and Jean-François Monin**
Subject : Proxy-based small inversions for the Rocq prover.
UGA, Verimag, France. To be defended the 26th of November, 2026.

Publications

October 2026 **A Minimalist Approach to Trustworthy Programming with Precise Types using Small Inversions**
Pierre Corbineau, Basile Gros, Jean-François Monin
In Erika Abraham; Nikolaj Bjorner; Michael Rawson, LPAR 2026, Spetses, Greece

September 2025 **Certified Programming with Dependent Types Made Simple with Proxy-based Small Inversions**
Pierre Corbineau, Basile Gros, Jean-François Monin
In Pierre Boutry; Loïc Pujet, the Rocqshop 2025, Reykjavik, Iceland

January 2025 **Proxy-based small inversions: a case study in MetaCoq programming**
Basile Gros, Pierre Corbineau, Jean-François Monin
In Adrien Guatto and Marie Kerjean, editors, JFLA 2025 - 36es Journées Francophones des Langages Applicatifs.

Teaching

2023-2025 **Introduction to logic and automated proving**
Teaching assistant for two consecutive years, L2, DLST UGA

Events Organised

2026 **Non-permanent day at VERIMAG laboratory**
A day-long event where PhD student, postdocs, and interns presented their work in front of the whole laboratory.

Academic Formation

Diplomas

- 2023 **Master's Degree in Computer Science : Performance in Software, Media and Scientific Computing**
dual degree, ENSEEIHT, Toulouse, France
- 2023 **Engineer's Degree in Computer Science**
dual degree, ENSEEIHT, Toulouse, France

Studies

- 2020 - 2023 **Engineering course on Computer Science, Software Systems specialty**
ENSEEIHT, Toulouse, France
- 2018 - 2020 **Classe Préparatoire aux Grandes Ecoles MPSI-MP***
Lycee Champollion, Grenoble, France

Internships for my Engineer's and Master's degrees

- 2023 18-week research internship on integrating user-side instructions to direct compiler optimisation in a toy language (supervised by Pierre-Evariste Dagand at IRIF, Paris)
- 2022 12-week research internship on formalizing category theory theorems in AGDA (supervised by Tarmo Uustalu at Reykjavik University, Iceland)
- 2021 8-week internship on working on a serious game about satellites (GarageSat) (supervised by Ramon Gomez-Moya, Federation Open Space Maker)